



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.

Department of Computer Science & Engineering

Academic Year: 2025-26 (semester II)

In-Semester Examination II

Class: TY B.Tech.

Course Name: Unix System
Programming

Course Code: PCC-CS312

Date: 29/03/2026

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.

Q.N.		Marks	CO	BT
Q.1	Draw the state transition diagram of a Unix process. What is zombie state?	7	CO3	BT1
Q.2	What is difference between pipe() and Popen()?	3	CO1	BT2
Q.3	Explain how pipes works by drawing a neat diagram.	5	CO1	BT2
Q.4	Write a simple code to show how fork() and wait() can be used.	5	CO3	BT3

=== ALL THE BEST ===



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science and Engineering
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: TY CSE

**Course Name: : Elective : Cyber
Security / Elective: Fundamentals of Cyber
Security**

**Course Code: PEC-
CS314/PEC-CS304**

Date: 23.03.2026

Time: 11:00AM

Max. Marks: 20

Note:

i. All questions are compulsory

Q.N.			Marks	CO	BT
Q.1		Explain the Three-Party Key Protocol (3KP) in detail with a neat message flow. Also differentiate between Two-Party Key Protocol (2KP) and Three-Party Key Protocol (3KP) used in digital transaction systems.	10	CO 1, 2,3	BT1 BT2
Q.2		Explain the design and working mechanism of the Unified Payments Interface (UPI) in detail. Describe its overall architecture, key components such as payer, payee, payment service providers (PSPs), banks, and the role of the National Payments Corporation of India (NPCI) in transaction processing. Also explain the step-by-step flow of a UPI transaction, including authentication, fund transfer, and confirmation, with the help of a neat diagram.	10	CO1 ,2, 3	BT2 BT3 BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT)



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Computer Science & Engineering
End Semester Examination
Academic Year: 2025-26 (Semester –II)

Class: TY B. Tech.	Subject: Information Retrieval	Subject Code: PEC-CS306
Date: 23/03/2026	Time: 11:00-11:45	Max. Marks: 20

Note:

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Q.N.	Solve all Questions	Marks	CO	BT
Q.1)	What is Hierarchical Clustering? Explain difference between Agglomerative and Divisive Clustering	5	CO3	BT2
Q.2)	Explain linkage clustering methods Single Linkage, Complete Linkage, Average Linkage, and Centroid Linkage in detail with diagram.	10	CO2	BT3
Q.3)	Document Terms D1 web, search D2 search, engine D3 web, engine D4 web, search, engine Evaluate Query: (web AND search) OR engine using Boolean retrieval model.	5	CO2	BT3

Bloom's Taxonomy (BT)



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Academic Year: 2025-2026

Semester: II

Examination: ISE 2

Class: T.Y. (CSE)

Branch: CSE

Date: 24/03/2026

Time: 11:00 To 11:45

Subject: Advanced Algorithms

Code: PEC-CS312/
PCC-CS313

Max. Marks: 20

Solve All Questions. Figures to right indicate full marks.

Marks

Q.1. Explain the importance of Red-Black tree in data structures. Discuss in detail the different cases of insert and delete operations of it. Assume initially the Red-Black tree is empty, and perform the following operations:

(10)

- Given the following sequence of numbers: 14, 24, 34, 19, 29.
 - Insert these numbers into an initially empty Red-Black tree.
 - Show the tree structure after each insertion and indicate any rotations or recoloring that occur.
- Now, delete 24 and show the tree after the delete operation.

Q.2. An operating system uses a k -bit binary counter to track the number of page faults that occur during a process's execution. The counter is initialized to 0. The only standard operation is INCREMENT, which adds 1 to the binary value. The actual cost of a single INCREMENT operation is exactly equal to the number of bits that must be flipped to achieve the new value.

(10)

- Define the Potential Function to be used for Amortized Analysis.
- Do the Amortized Cost Calculation for each INCREMENT operation and Total Amortized Cost after last INCREMENT operation.
- Suppose the system designers introduce a RESET operation that forcefully clears the entire k -bit array back to zeroes, with an actual execution cost of k (since it must iterate through and overwrite every bit). If the counter is subjected to an arbitrary sequence of INCREMENT and RESET operations, does the amortized cost per operation remain $O(1)$? Justify your answer using the potential function from Part A answer.

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S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
ISE-I Examination Sem-II, AY 2025-26
Class: Third Year (Computer Science and Engineering)

Date: 28/03/2026

Time: 04:30 to 05:15

Subject: Compiler Design
(PCC-CS302/PCC-CS310)

Max. Marks: 20

Note: - Solve sequentially and assume suitable data if required.

Q. No.	Solve the following.	Marks	CO(BT)
1	Remove left recursion from the grammar: $S \rightarrow A a \mid b$ $A \rightarrow S b \mid A c \mid d$	3	CO2(BT2)
2	Remove ϵ -productions from the grammar: $S \rightarrow AB \mid C$ $A \rightarrow aA \mid \epsilon$ $B \rightarrow bB \mid \epsilon$ $C \rightarrow c$	3	CO2(BT3)
3	Left factored the give grammar. $S \rightarrow aSbSc \mid aSbSd \mid aSe \mid f$	3	CO2(BT3)
4	$S \rightarrow aSbS \mid bSaS \mid \epsilon$ Construct FIRST & FOLLOW sets and LL(1) parsing table.	3	CO3(BT4)
5	For the grammar: $S \rightarrow (S)S \mid \epsilon$ Construct a parsing table and parse the string: "(())" using an SLR(1) parser.	8	CO3(BT5)

*****All the Best*****



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science & Engineering
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: TY **Course Name: Digital Forensic** **Course Code: MDM-IS-CS302**
Date: 29.03.2026 **Time: 11:00 to 11:45** **Max. Marks: 20**

Note:

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- iii. *Figures to right indicate full marks.*

Q.N.	Attempt All the Questions.	Marks	CO	BT
Q.1	Explain the OWASP Top 10 security risk in detail with preventive code.	10	CO4	BT3
Q.2	Write a short note on windows forensic volatile information.	05	CO3	BT1
Q.3	Write a short note on password cracking using Rainbow Table.	05	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of XXXXXXXXXXXX
Academic Year: 2025-26 (semester II)
In-Semester Examination II

Class: TY B.Tech	Course Name: Software Engineering	Course Code: MDM-CS302
Date: 29.03.2026	Time: 11.00	Max. Marks: 20

Note:

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- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Explain phases of Project Management Process	05	CO2	BT1
	b	Explain User Interface Design and its type	05	CO3	BT2
Q.2	a	Explain the different types of coupling	05	CO4	BT3
	b	What is the Information System Life Cycle (ISLC)? Explain	05	CO4	BT5

=== ALL THE BEST ===



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science & engineering.

Academic Year: 2025-26

In Semester Examination -2 (ISE-2)

Course Name: Database Management Systems

Time: 11.00

Course Code: PCC-CS304

Max. Marks: 20

Class: Third Year .

Date: 25.03.2026

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

computer

Q.N.		Marks	CO	BT
Q.1.	What is Functional Dependency? State and explain different axioms for computing closure of Functional dependency. Consider the following set F of functional dependencies for relation schema $R = (A, B, C, D, E)$ d) Compute the closure of F i.e F^+ e) List the candidate keys for R. f) Compute the canonical cover F_c. $F = \{ A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A \}$	08	CO4	BT3
Q.2.	(a) What are aggregate functions? Discuss with examples different aggregate functions. (b) Explain the distinctions among the terms primary key, candidate key, and super key.	06	CO3	BT1
Q.3.	Construct an E-R diagram for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted	06	CO2	BT2